

## JIULONG, China

WMO: 564620

Lat: 29.0080N

Lon: 101.5009E

Elev: 2931

StdP: 70.73

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.7       | -5.6 | -21.7                         | 0.8 | 7.9  | -19.6 | 0.9 | 6.8  | 8.8                   | 7.6  | 7.6 | 7.5  | 1.9                   | 320  | 0.551 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.0                    | 25.3            | 13.2 | 24.0 | 12.9 | 22.7 | 12.6 | 15.7                | 21.0 | 15.2 | 20.1 | 14.8 | 19.4 | 5.7                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.3                            | 14.7 | 16.6 | 13.9 | 14.3 | 16.3 | 13.5 | 13.9 | 16.1 | 56.3          | 21.1 | 54.5 | 20.2 | 53.0 | 19.5 | 22.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 8.4               | 7.4  | 6.4 | -10.0                      | 27.8  | 1.1                | 1.2 | -10.8                                              | 28.7  | -11.4      | 29.4  | -12.0      | 30.0  | -12.8      | 30.9  |      |
|                   |      |     | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |      |
|                   |      |     | WB                         | -11.7 | 17.1               | 0.9 | 1.5                                                | -12.3 | 18.2       | -12.9 | 19.1       | -13.3 | 20.0       | -14.0 | 21.1 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.3   | 2.6   | 5.0   | 7.6   | 10.6  | 13.8  | 15.8  | 16.4  | 16.2  | 14.6  | 11.4  | 6.4   | 3.0  |
|                                                                                  | DBStd       | 5.49   | 1.90  | 2.33  | 2.35  | 2.47  | 2.88  | 2.47  | 1.86  | 2.01  | 2.42  | 2.54  | 2.05  | 1.88 |
|                                                                                  | HDD10.0     | 817    | 228   | 140   | 80    | 21    | 5     | 1     | 0     | 0     | 2     | 15    | 109   | 217  |
|                                                                                  | HDD18.3     | 2938   | 486   | 373   | 333   | 232   | 142   | 80    | 63    | 69    | 112   | 215   | 358   | 475  |
|                                                                                  | CDD10.0     | 936    | 0     | 0     | 5     | 40    | 123   | 176   | 199   | 193   | 141   | 58    | 1     | 0    |
|                                                                                  | CDD18.3     | 17     | 0     | 0     | 0     | 0     | 2     | 6     | 4     | 4     | 2     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 187    | 0     | 0     | 1     | 7     | 48    | 49    | 32    | 41    | 10    | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 9      | 0     | 0     | 0     | 0     | 3     | 4     | 1     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 2.4   | 2.7   | 2.8   | 2.9   | 2.9   | 2.5   | 2.3   | 2.3   | 2.1   | 2.3   | 2.4   | 2.4  |
| Precipitation                                                                    | PrecAvg     | 907    | 2     | 4     | 12    | 44    | 91    | 197   | 185   | 136   | 161   | 63    | 8     | 3    |
|                                                                                  | PrecMax     | 1218   | 10    | 19    | 40    | 114   | 190   | 347   | 277   | 268   | 258   | 137   | 20    | 10   |
|                                                                                  | PrecMin     | 678    | 0     | 0     | 1     | 14    | 24    | 87    | 113   | 39    | 88    | 7     | 0     | 0    |
|                                                                                  | PrecStd     | 132    | 2     | 4     | 8     | 22    | 42    | 57    | 39    | 55    | 40    | 32    | 5     | 3    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.3  | 19.7  | 21.9  | 24.4  | 26.8  | 27.2  | 25.8  | 26.3  | 24.9  | 21.6  | 17.3  | 13.6 |
|                                                                                  |             | MCWB   | 2.8   | 4.5   | 6.5   | 8.4   | 12.0  | 14.4  | 14.6  | 14.1  | 13.8  | 11.7  | 6.6   | 3.2  |
|                                                                                  | 2%          | DB     | 13.0  | 16.9  | 19.3  | 22.1  | 24.7  | 24.6  | 24.1  | 24.5  | 22.6  | 19.8  | 15.4  | 12.1 |
|                                                                                  |             | MCWB   | 1.8   | 3.7   | 5.5   | 7.7   | 11.3  | 13.8  | 14.4  | 13.8  | 13.5  | 10.8  | 5.7   | 2.4  |
|                                                                                  | 5%          | DB     | 11.2  | 14.8  | 17.4  | 20.4  | 23.0  | 22.9  | 22.5  | 22.8  | 20.9  | 18.3  | 14.0  | 10.7 |
|                                                                                  |             | MCWB   | 1.1   | 2.9   | 4.7   | 7.3   | 10.5  | 13.8  | 14.4  | 13.8  | 13.2  | 10.4  | 5.2   | 1.8  |
|                                                                                  | 10%         | DB     | 9.5   | 12.8  | 15.5  | 18.7  | 21.2  | 21.2  | 20.9  | 21.1  | 19.3  | 16.9  | 12.4  | 9.3  |
|                                                                                  |             | MCWB   | 0.3   | 2.1   | 4.1   | 6.9   | 10.3  | 13.7  | 14.2  | 13.7  | 13.0  | 9.8   | 4.5   | 1.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.9   | 5.3   | 7.7   | 10.4  | 13.8  | 16.3  | 16.6  | 15.9  | 15.2  | 13.2  | 8.5   | 4.4  |
|                                                                                  |             | MCDB   | 12.2  | 16.3  | 18.0  | 19.5  | 21.9  | 23.0  | 22.1  | 21.5  | 21.3  | 18.5  | 13.6  | 11.8 |
|                                                                                  | 2%          | WB     | 2.6   | 4.2   | 6.4   | 9.2   | 13.0  | 15.5  | 15.8  | 15.2  | 14.6  | 12.5  | 7.1   | 3.2  |
|                                                                                  |             | MCDB   | 11.3  | 15.2  | 15.8  | 17.7  | 20.4  | 21.3  | 20.7  | 20.4  | 20.2  | 17.2  | 13.0  | 10.3 |
|                                                                                  | 5%          | WB     | 1.5   | 3.3   | 5.6   | 8.5   | 12.3  | 14.9  | 15.3  | 14.8  | 14.1  | 11.8  | 6.2   | 2.4  |
|                                                                                  |             | MCDB   | 9.9   | 13.4  | 14.6  | 16.7  | 19.0  | 19.9  | 19.8  | 19.6  | 19.0  | 16.0  | 12.0  | 9.3  |
|                                                                                  | 10%         | WB     | 0.6   | 2.4   | 4.8   | 7.8   | 11.6  | 14.4  | 14.9  | 14.3  | 13.6  | 10.9  | 5.3   | 1.5  |
|                                                                                  |             | MCDB   | 8.6   | 11.7  | 13.3  | 15.8  | 17.8  | 18.9  | 18.9  | 18.8  | 17.8  | 14.8  | 11.0  | 8.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.4  | 16.2  | 15.4  | 14.9  | 12.5  | 9.8   | 9.0   | 9.3   | 8.9   | 10.9  | 13.5  | 14.6 |
|                                                                                  |             | MCDBR  | 17.9  | 19.7  | 19.2  | 18.6  | 16.6  | 12.9  | 12.1  | 12.8  | 11.6  | 12.7  | 15.8  | 16.1 |
|                                                                                  |             | MCWBR  | 9.0   | 9.2   | 8.1   | 7.0   | 5.1   | 3.9   | 3.9   | 3.7   | 3.8   | 4.7   | 6.9   | 7.8  |
|                                                                                  | 5% WB       | MCDBR  | 16.4  | 18.5  | 16.8  | 15.4  | 11.9  | 10.2  | 9.0   | 9.2   | 9.1   | 9.7   | 13.2  | 14.7 |
|                                                                                  |             | MCWBR  | 8.4   | 8.7   | 7.0   | 5.9   | 3.9   | 3.7   | 3.2   | 3.0   | 3.2   | 3.8   | 6.1   | 7.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.254  | 0.282 | 0.365 | 0.390 | 0.378 | 0.377 | 0.392 | 0.399 | 0.367 | 0.313 | 0.267 | 0.245 |      |
|                                                                                  | taud        | 2.535  | 2.470 | 2.211 | 2.190 | 2.308 | 2.394 | 2.350 | 2.312 | 2.384 | 2.502 | 2.568 | 2.589 |      |
|                                                                                  | Ebn at Noon | 1005   | 995   | 920   | 905   | 911   | 907   | 892   | 885   | 907   | 947   | 981   | 1002  |      |
|                                                                                  | Edh at Noon | 86     | 101   | 140   | 148   | 132   | 121   | 126   | 129   | 116   | 96    | 83    | 78    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.06   | 4.65  | 5.04  | 5.24  | 5.42  | 4.84  | 4.58  | 4.51  | 4.18  | 4.12  | 4.00  | 3.86  |      |
|                                                                                  | RadStd      | 0.26   | 0.31  | 0.45  | 0.51  | 0.37  | 0.47  | 0.36  | 0.54  | 0.31  | 0.32  | 0.36  | 0.21  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) | Station Only           | +0.35 | N/A     | N/A    | +0.68   | +0.28 | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (6 neighbors) | +0.35 | N/A     | -0.51  | +0.58   | N/A   | N/A         | N/A     | -92     | +190    |

Nomenclature: See separate page